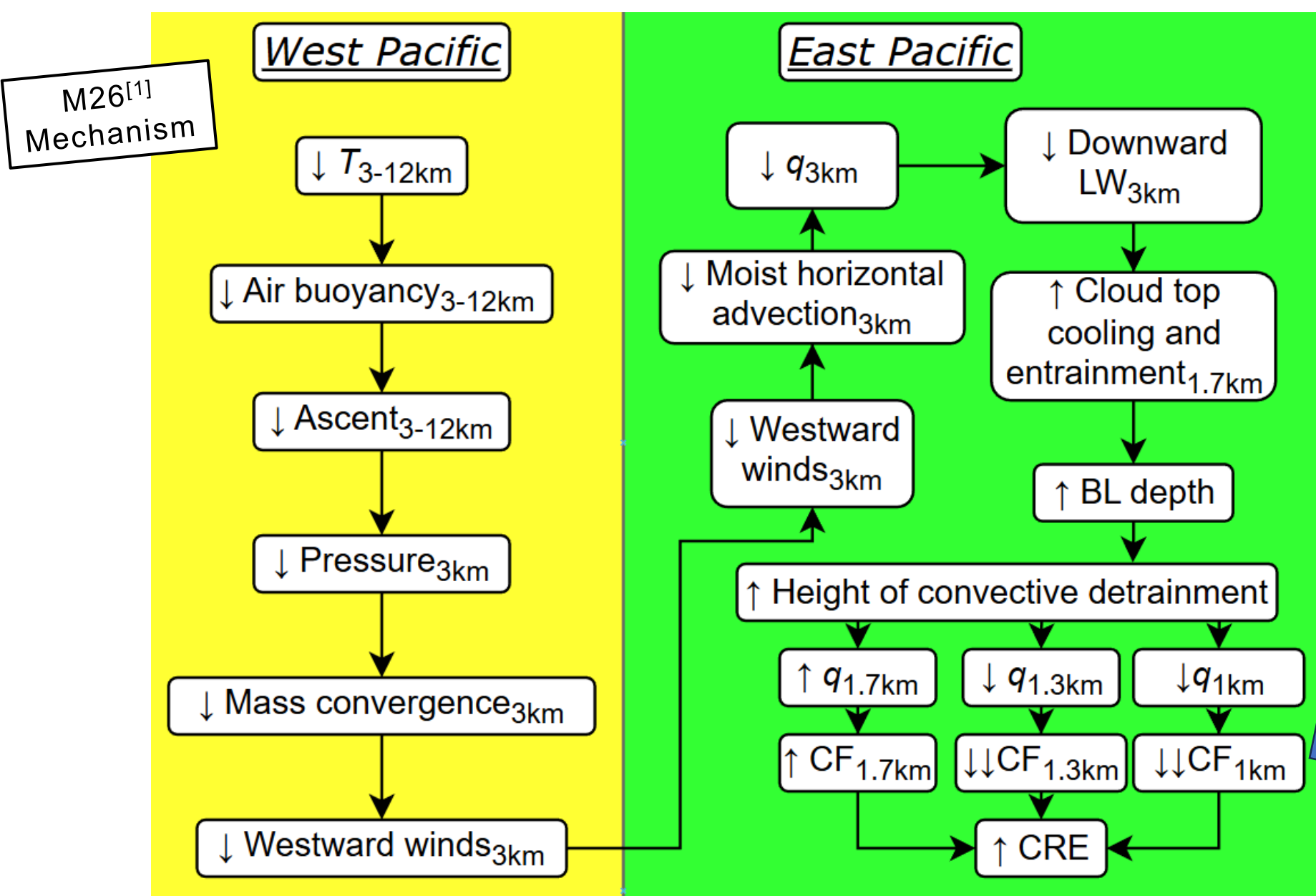


1-sentence summary: The West Pacific (WP) modulates Low Clouds (LC) in the East Pacific (EP) by equatorial advection **and** subtropical Rossby waves

Background

- Low clouds (LC) are one of the largest sources of climate uncertainty in climate models [1,2]
- How large-scale circulation affects LC within models is poorly understood
- McCulloch *et al.*, 2026^[1] (M26) proposed the following mechanism for WP→EP interaction:



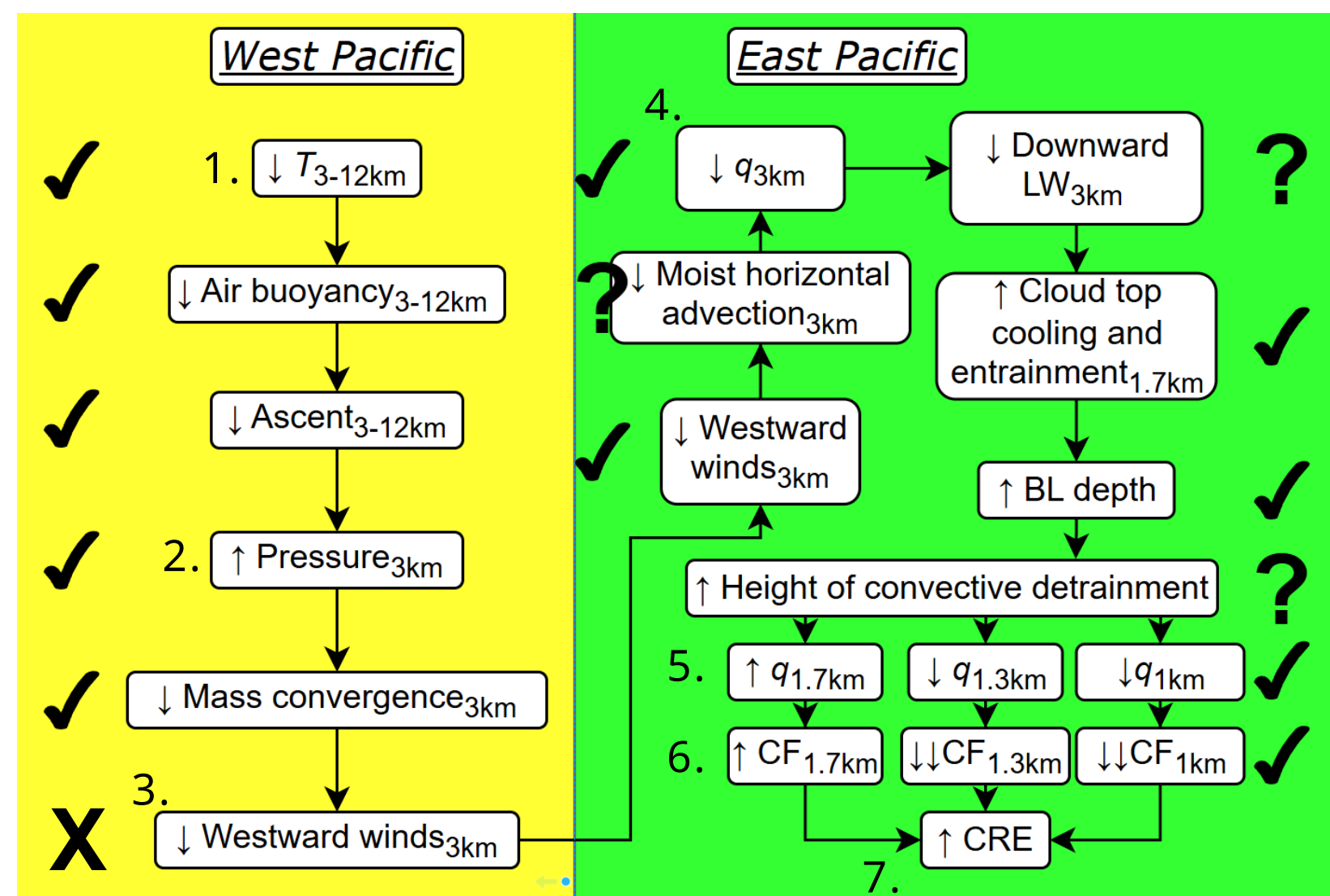
- In this work we test this mechanism by applying an identical forcing over short time scales and observing the immediate response

What would you like to see next?

- Detailed results over the 60-days
- Overview of the two pathways of how the WP effects the EP
- How the perturbation is applied

Key results

- Main result: The signal is noisy, but anomaly patterns in-line with the 20-year average seen in M26 do emerge after 60 days
- Results for statistically significant anomalies are shown for days 41:60 (numbers below denotes which plot)
- 1. The WP T , q and u responses are as seen in M26^[1]
- 2. $\uparrow$ Pressure_{3km} does occur across the equatorial Pacific
- 3. $\downarrow$ Westward winds_{3km} across the Pacific are not seen (but u response is very noisy)
- 4. $\downarrow$ q_{3km} does occur across the equatorial Pacific
- 5. $\Delta q_{1.7}$ & $1.3km$ is noisy, but does occur in the EP
- 6. $\downarrow\downarrow CF_{1.3km}$ and $\uparrow CF_{1.7km}$ are consistent with M26
- 7. $\uparrow CRE_{net}$ in the EP is noisy but consistent with M26
- Where the response agrees (✓) or disagrees (✗) with the M26 mechanism is shown below
- Some variables were untested due to computational constraints (shown with ?)



- 8. ΔT_{13km} shows a Rossby wave response propagating from the Southern Pacific into EP_{bot}. This likely drives stability changes in EP_{bot}

Main finding: Stability in EP_{bot} likely affected by Rossby wave. EP_{top} still driven by M26 mechanism

[1] McCulloch, D., Webb, M. J., Lambert, F. H., & Vallis, G. K. (2026). Weakening tropical West Pacific ascent strength reduces East Pacific low clouds via lower free-tropospheric moisture convergence. *Journal of Advances in Modeling Earth Systems*, 18, <https://doi.org/10.1029/2025MS005386>

[2] Sherwood, S. C., Webb, M. J., Annan, J. D., Armour, K. C., Forster, P. M., Hargreaves, J. C., et al. (2020). An assessment of Earth's climate sensitivity using multiple lines of evidence. *Reviews of Geophysics*, 58, e2019RG000678. <https://doi.org/10.1029/2019RG000678>



Model:

- We use the Met Office Unified Model (UM) in the following configuration:
- Atmosphere-only AMIP suite between 1985 and 2005 (with prescribed SST's)
- Global Land and Atmosphere 9 (GA9/GL9) configurations
- N96 resolution with 85km model top height

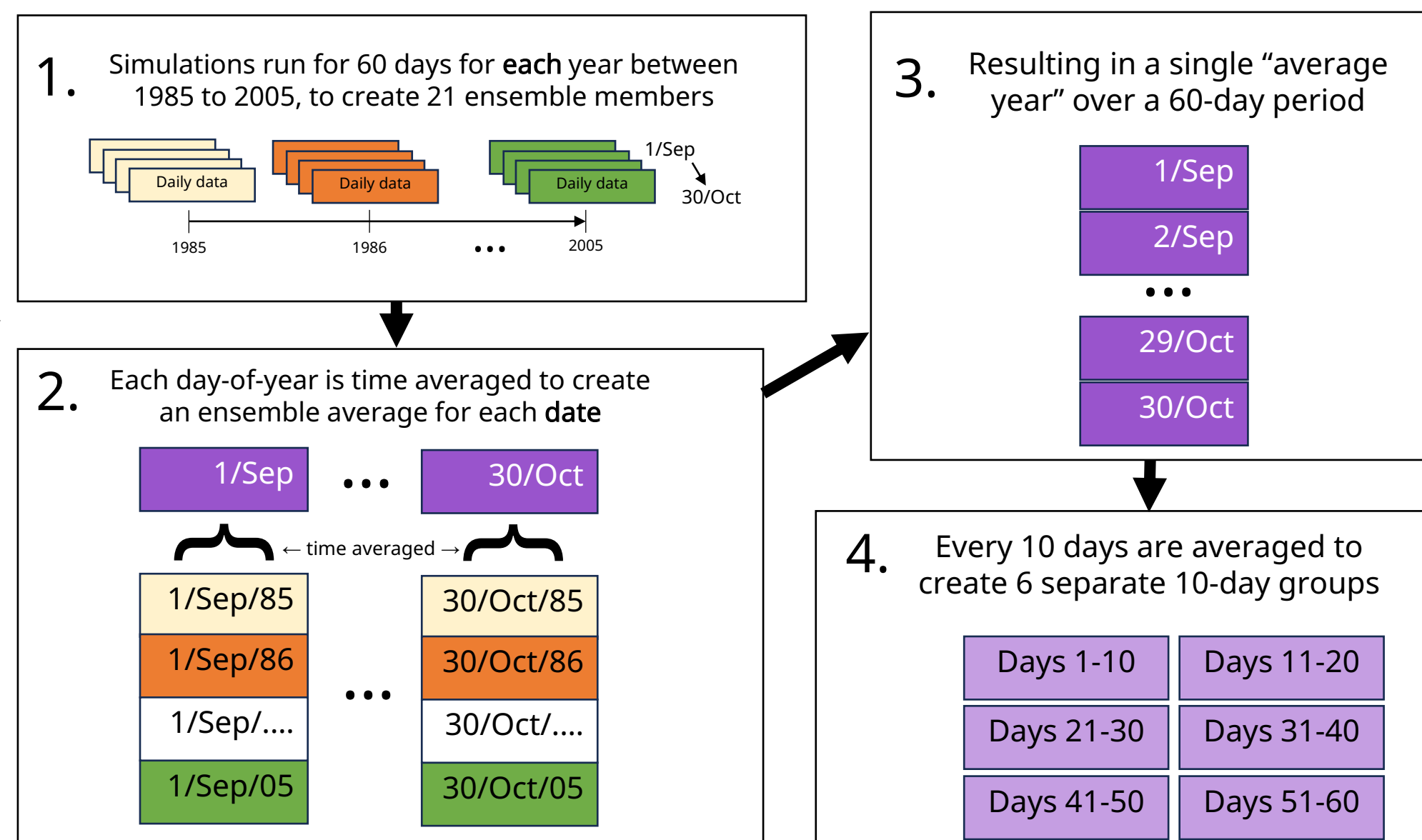
Perturbation:

- A cooling perturbation of 0.648 K/day (0.009 K/timestep) is applied between 3 and 12 km in the WP (yellow box in the map below)
- The first 60 days are outputted and processed using this method
- A Welch's t -test is done on the 10-day groups and only values exceeding a 90% confidence threshold are shown (to filter noise)

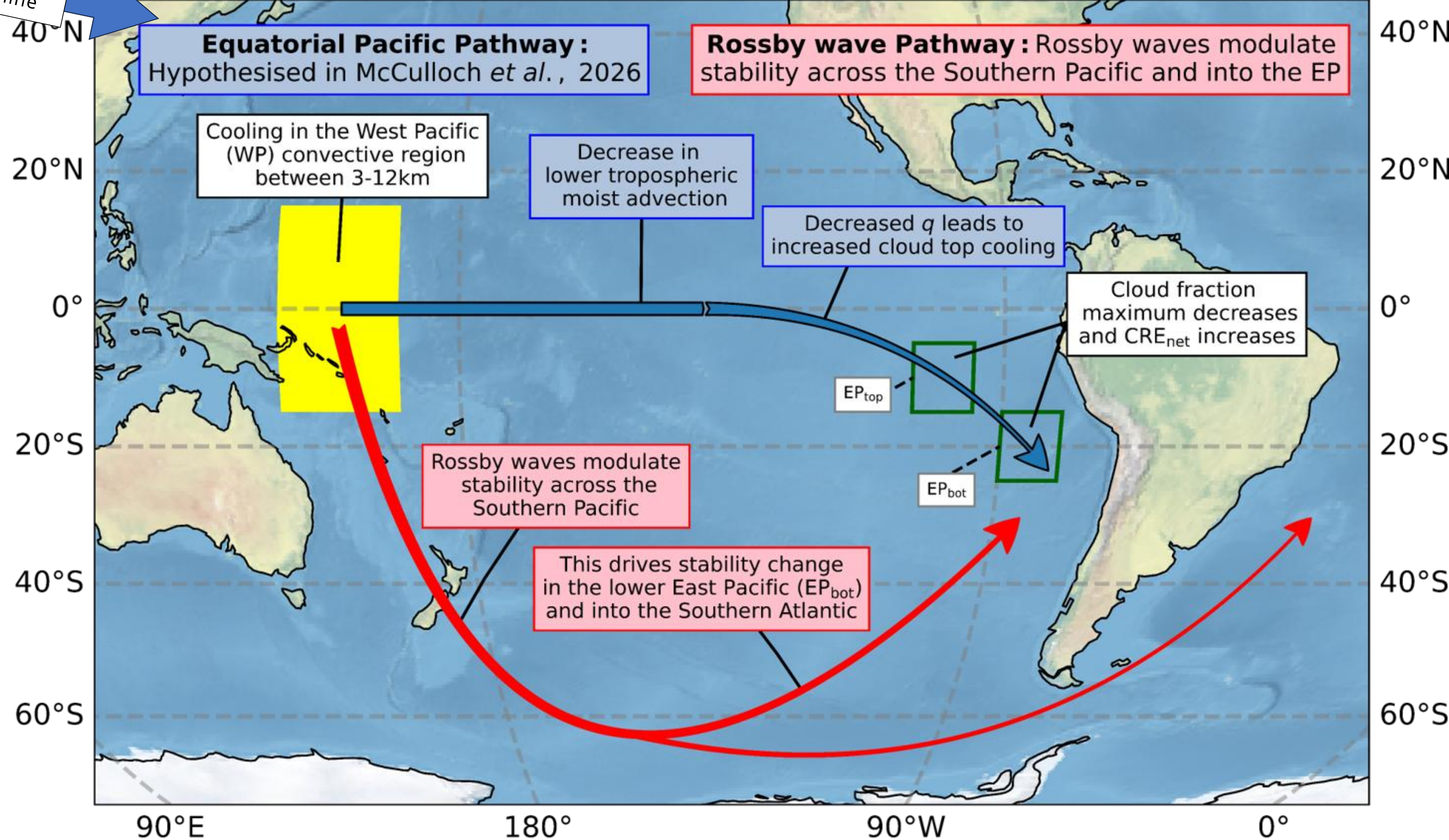
Methods

Temporal averaging and technique

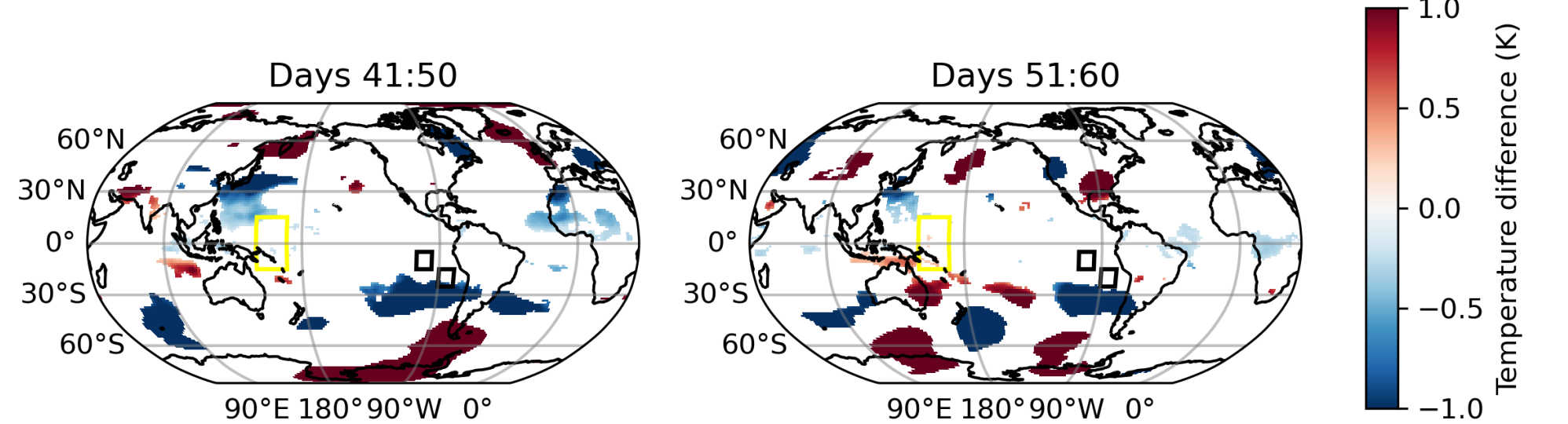
- The perturbation is applied across 21 ensemble members using the technique outlined below
- The first 60 days of the forcing are binned into 10-day groups, only days 41:60 are shown here



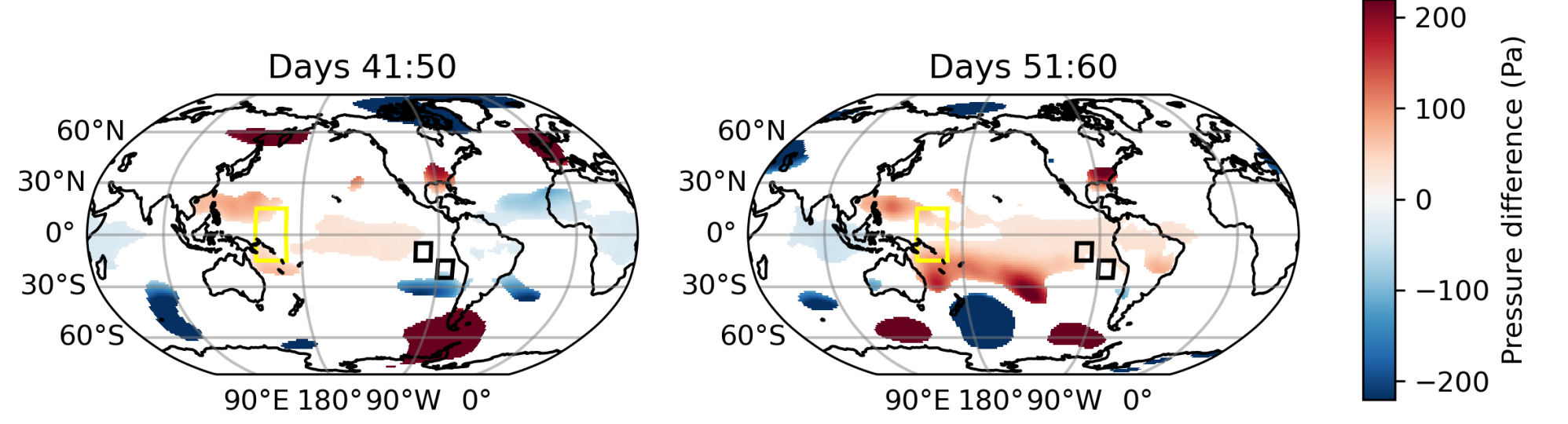
How the West Pacific affects the East Pacific



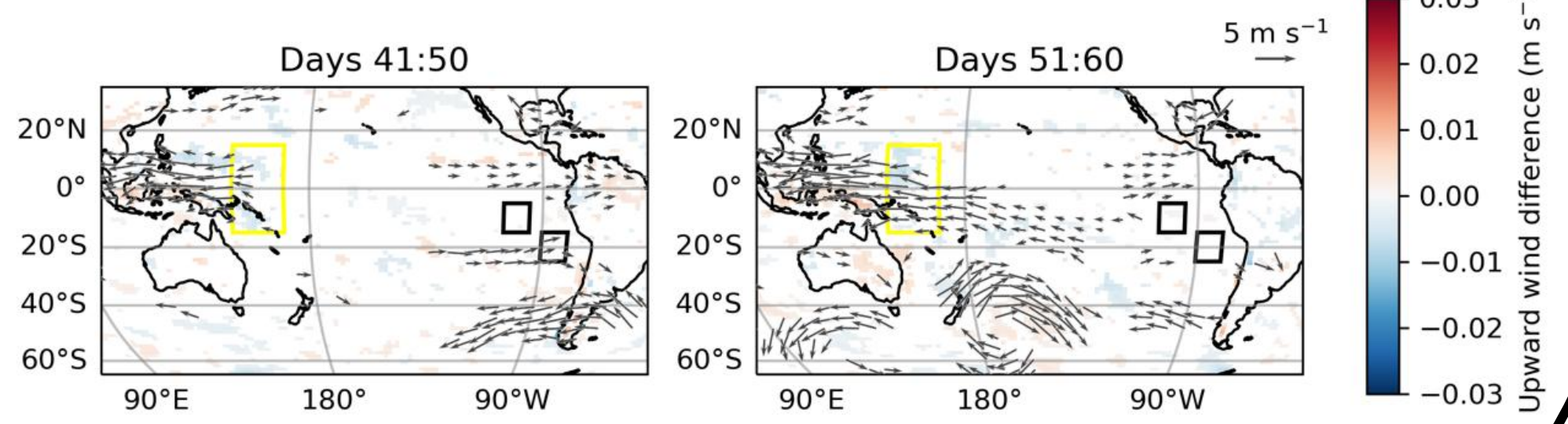
1. Air temperature difference at 3km | 10-day periods



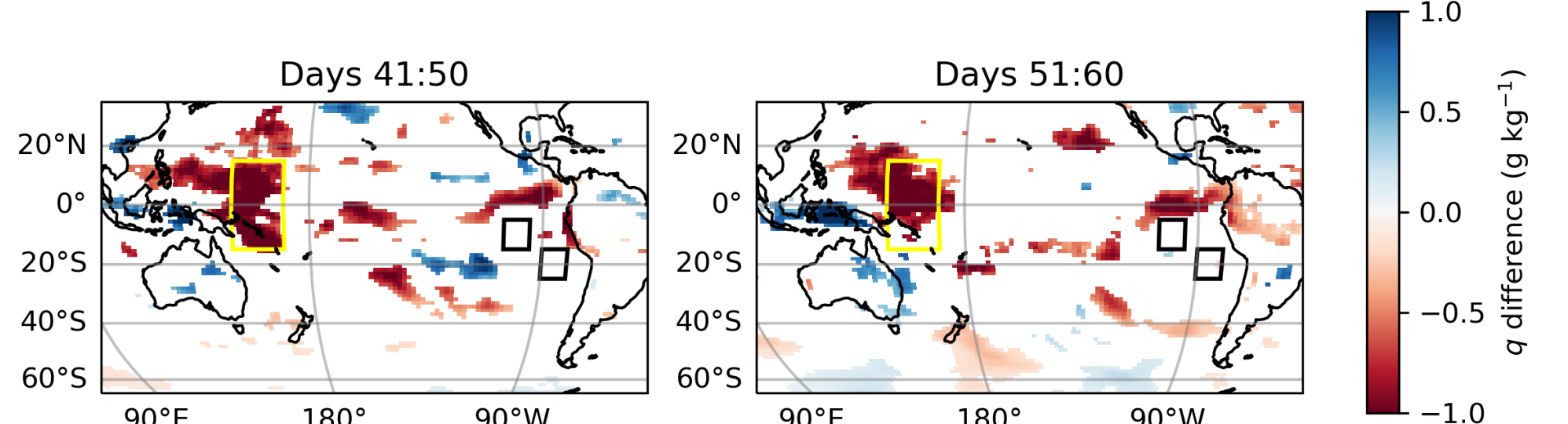
2. Air pressure difference at 3km | 10-day periods



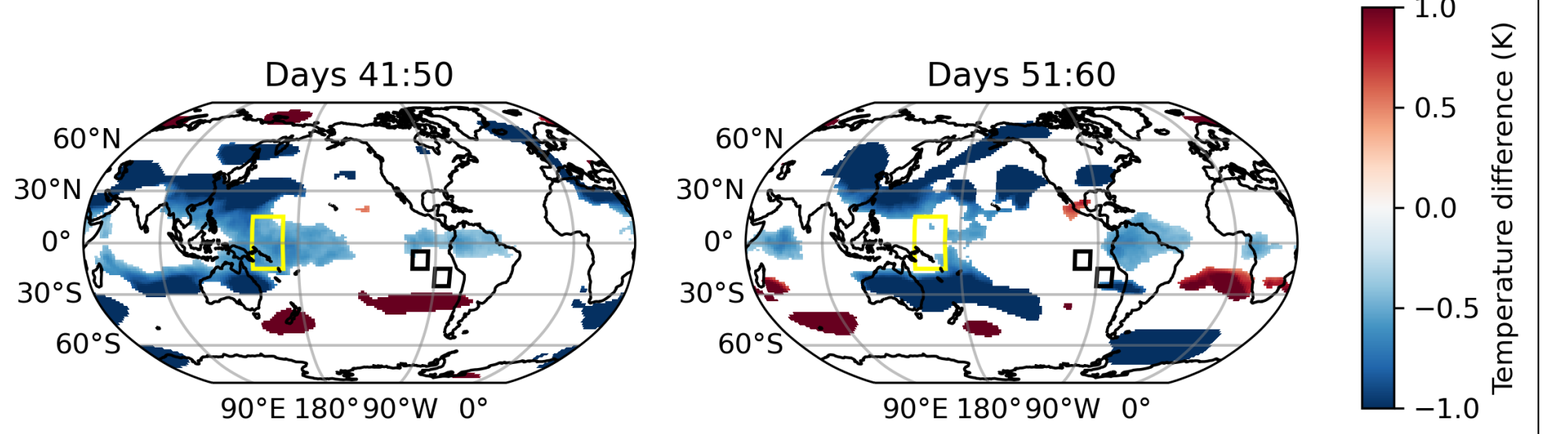
3. Wind differences at 3km | 10-day periods



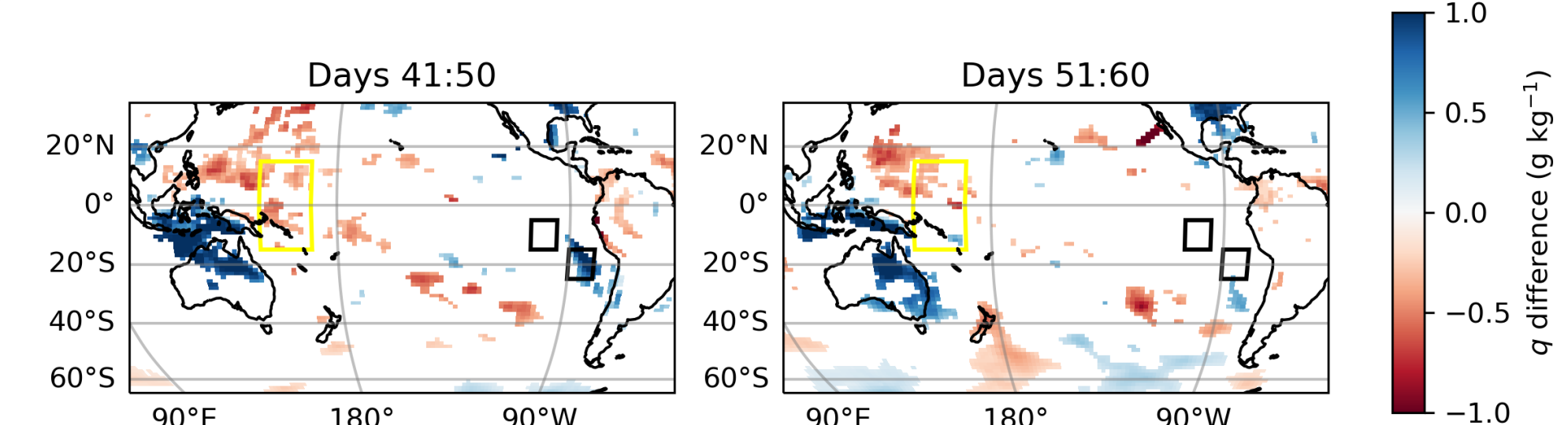
4. Specific humidity difference at 3km | 10-day periods



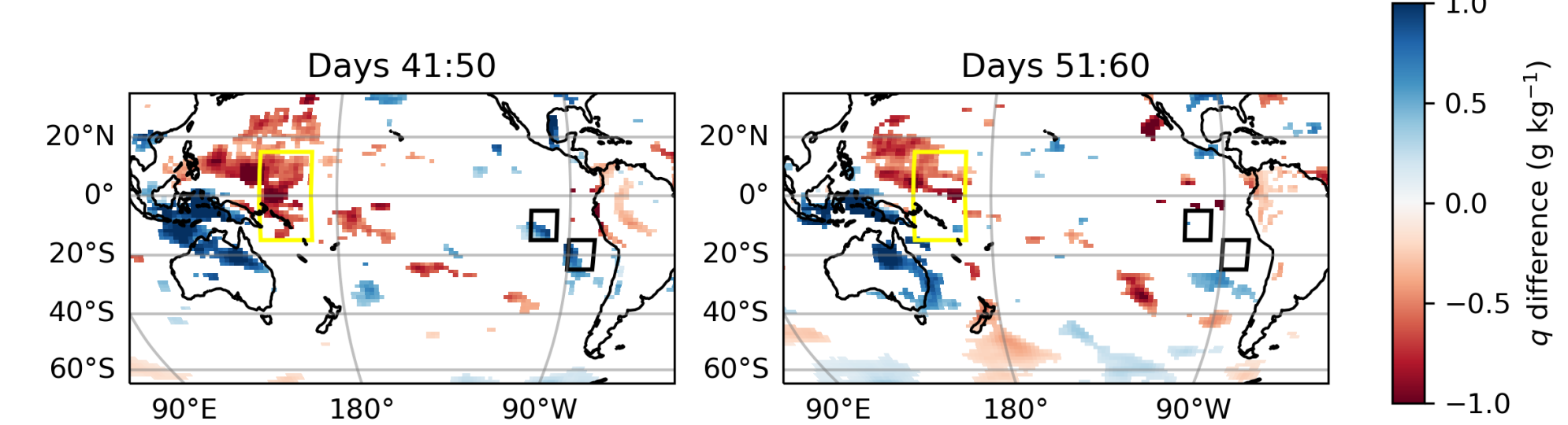
8. Air temperature difference at 13km | 10-day periods



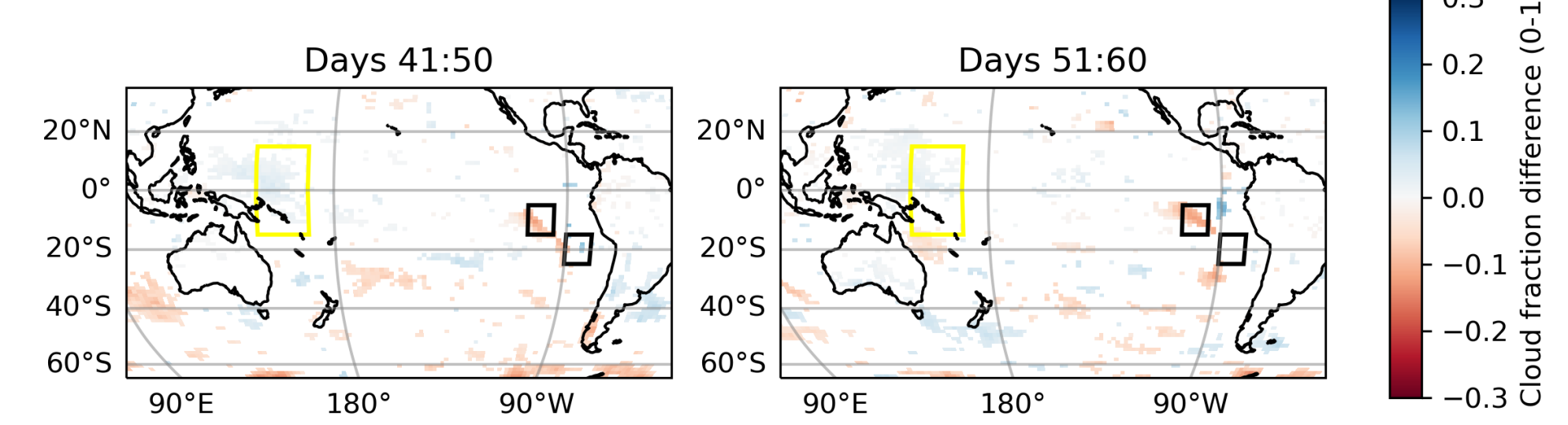
5a. Specific humidity difference at 1.3km | 10-day periods



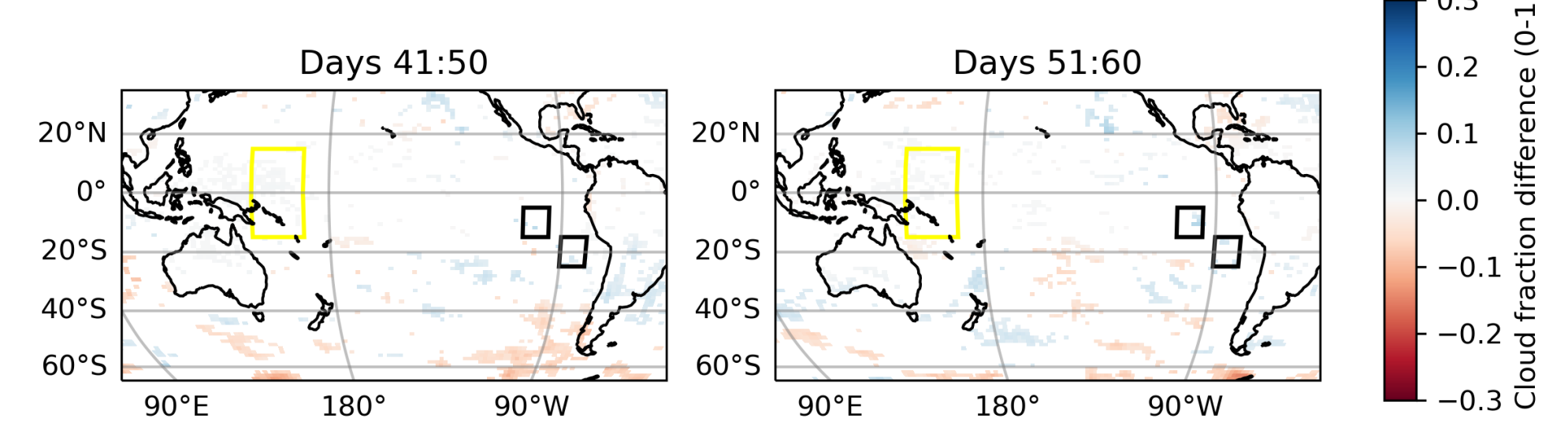
5b. Specific humidity difference at 1.7km | 10-day periods



6a. Cloud volume difference at 1.3km | 10-day periods



6b. Cloud volume difference at 1.7km | 10-day periods



7. CREnet differences | 10-day periods

